

Unit 1: Defining Cloud Computing

- Cloud Computing has fundamentally changed how businesses and individuals store, access, and manage data.
- It is a technology that allows you to use data and applications over the internet, moving beyond the limitations of a local computer's hard drive or a private server.

The Problem Cloud Computing Solves: The "Old Way" vs. The "New Way"

Before the cloud, if a company wanted to launch a website or an application, it had to follow a slow and expensive process:

- **The "Old Way" (On-Premises):** The company had to buy powerful physical servers, storage disks, and networking equipment. This required a huge upfront investment, known as **Capital Expenditure (CapEx)**. They had to guess their peak traffic needs, often buying far more capacity than they used day-to-day, leading to wasted resources. If they needed a new server, the process of ordering, installing, and configuring it could take weeks or even months.
- **The "New Way" (Cloud Computing):** The cloud changes this model entirely. Instead of buying hardware, you rent computing power from a cloud provider (like AWS, Google, or Microsoft). This shifts the cost from a large upfront investment to a manageable monthly bill, known as an **Operational Expenditure (OpEx)**. This model eliminates guesswork, long waiting times, and wasted resources.

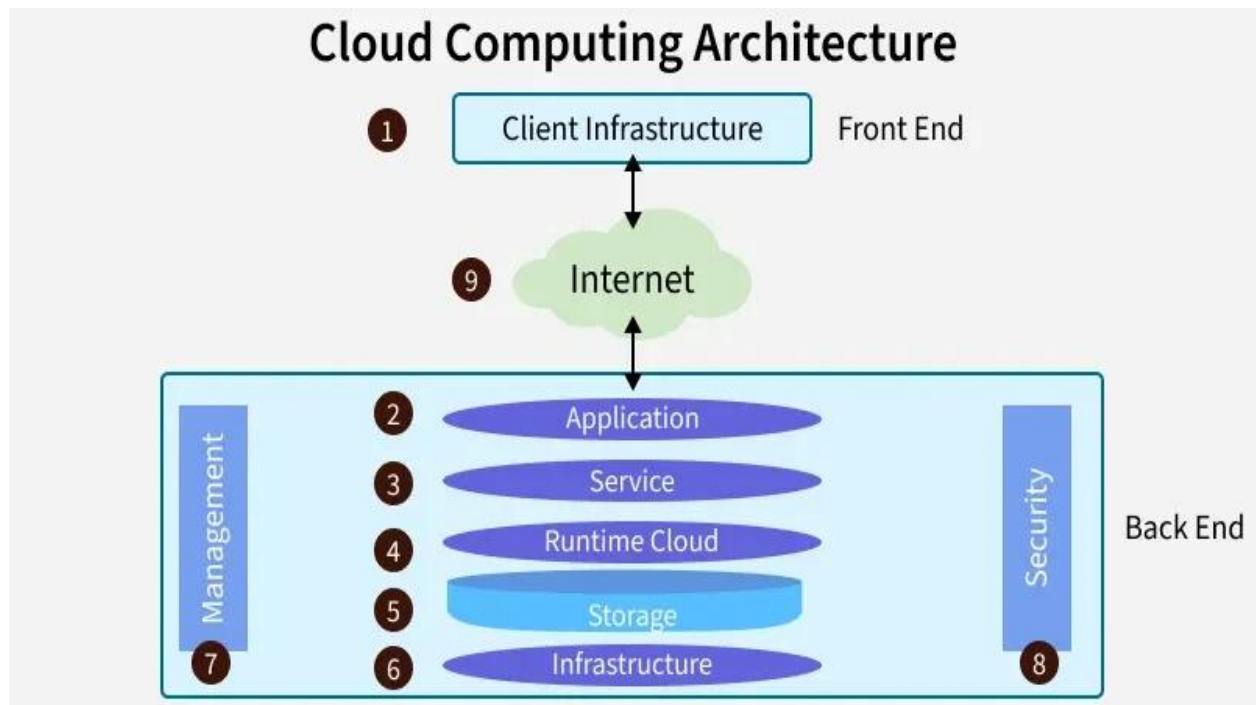
What is Cloud Computing?

Cloud Computing is a technology that allows you to store and access data and applications over the internet instead of using your computer's hard drive or a local server.

Architecture Of Cloud Computing

[Cloud computing architecture](#) refers to the components and sub-components required for cloud computing. These components typically refer to:

1. Front end (Fat client, Thin client)
2. Back-end platforms (Servers, Storage)
3. Cloud-based delivery and a network (Internet, Intranet, Intercloud)



1. Front End (User Interaction Enhancement)

The User Interface of Cloud Computing consists of 2 sections of clients. The Thin clients are the ones that use web browsers facilitating portable and lightweight accessibilities and others are known as Fat Clients that use many functionalities for offering a strong user experience.

2. Back-end Platforms (Cloud Computing Engine)

The core of cloud computing is made at back-end platforms with several servers for storage and processing computing. Management of Applications logic is managed through servers and effective data handling is provided by storage. The combination of these platforms at the backend offers the processing power, and capacity to manage and store data behind the cloud.

3. Cloud-Based Delivery and Network

On-demand access to the computer and resources is provided over the Internet, Intranet, and Intercloud. The Internet comes with global accessibility, the [Intranet](#) helps in internal communications of the services within the organization and the [Intercloud](#) enables interoperability across various cloud services. This dynamic network connectivity ensures an essential component of cloud computing architecture on guaranteeing easy access and data transfer.

The 5 Core Characteristics of Cloud Computing

All cloud services share five fundamental characteristics that define them:

1. **On-Demand Self-Service:** Users can provision computing resources like servers and storage automatically, without requiring human intervention from the service provider.
2. **Broad Network Access:** Capabilities are available over the network and can be accessed through standard mechanisms by any device, such as laptops, tablets, and mobile phones.
3. **Resource Pooling:** The provider's computing resources are pooled to serve multiple customers using a multi-tenant model. Resources are dynamically assigned and reassigned according to demand.
4. **Rapid Elasticity (Scalability):** Resources can be scaled up or down quickly and, in some cases, automatically, to meet demand. This ensures you have enough power during traffic spikes and aren't paying for idle resources during quiet periods.
5. **Measured Service (Pay-as-you-go Model):** Cloud systems automatically control and optimize resource use. Usage is monitored, controlled, and reported, providing transparency for both the provider and the consumer. You only pay for what you use.

The Shared Responsibility Model

Security in the cloud is a partnership between the cloud provider and the customer. This is known as the **Shared Responsibility Model**.

- **The Cloud Provider (e.g., AWS, Azure, GCP)** is responsible for the security *OF* the cloud. This includes the physical security of the data centers, the hardware, and the core networking infrastructure.
- **The Customer (You)** is responsible for security *IN* the cloud. This includes managing who has access to your resources, encrypting your data, configuring firewalls (like Network Policies), and securing your applications.

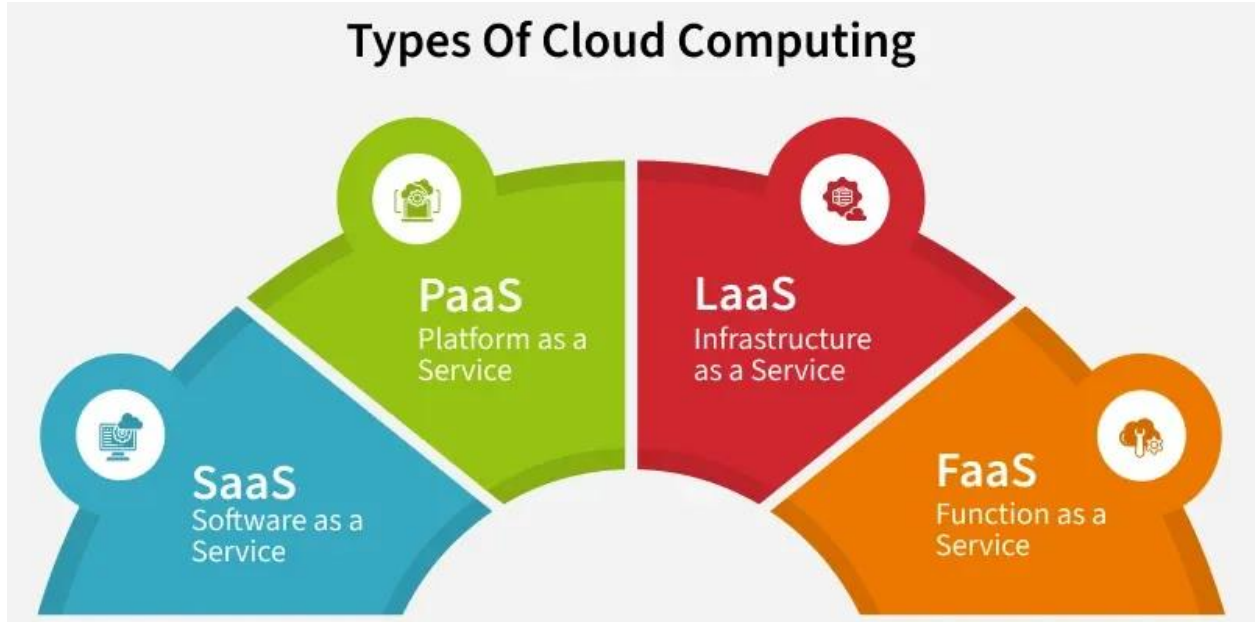
The amount of responsibility you have depends on the service model you choose.

Types of Cloud Computing Services

The following are the types of Cloud Computing:

1. **Infrastructure as a Service (IaaS)**
2. **Platform as a Service (PaaS)**

3. Software as a Service (SaaS)



1. Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) is a type of cloud computing that gives people access to IT tools like virtual computers, storage, and networks through the internet. You don't need to buy or manage physical hardware. Instead, you pay only for what you use.

Here are some key benefits of using IaaS:

- **Flexibility and Control:** IaaS comes up with providing virtualized computing resources such as VMs, Storage, and networks facilitating users with control over the Operating system and applications.
- **Reducing Expenses of Hardware:** IaaS provides business cost savings with the elimination of physical infrastructure investments making it cost-effective.
- **Scalability of Resources:** The cloud provides in scaling of hardware resources up or down as per demand facilitating optimal performance with cost efficiency.

2. Platform as a Service (PaaS)

Platform as a Service (PaaS) is a cloud computing model where a third-party provider offers the software and hardware tools needed to develop, test, and run applications. This allows users to focus on building their applications without worrying about managing servers or infrastructure.

For example, AWS Elastic Beanstalk is a PaaS offered by Amazon Web Services that helps developers quickly deploy and manage applications while AWS takes care of the needed resources like servers, load balancing, and scaling.

Here are some key benefits of using PaaS:

- **Simplifying the Development:** Platform as a Service offers application development by keeping the underlying Infrastructure as an Abstraction. It helps the developers to completely focus on application logic (Code) and background operations are completely managed by the AWS platform.
- **Enhancing Efficiency and Productivity:** PaaS lowers the Management of Infrastructure complexity, speeding up the Execution time and bringing the updates quickly to market by streamlining the development process.
- **Automation of Scaling:** Management of resource scaling, guaranteeing the program's workload efficiency is ensured by PaaS.

3. Software as a Service (SaaS)

Software as a Service (SaaS) is a way of using software over the internet instead of installing it on your computer. The software is hosted by a company, and you can use it just by logging in through a web browser. You don't need to worry about updates, maintenance, or storage the provider takes care of all that.

A common example is Google Docs. You can write and share documents online without downloading any software.

Here are some key benefits of using SaaS:

- **Collaboration And Accessibility:** Software as a Service (SaaS) helps users to easily access applications without having the requirement of local installations. It is fully managed by the AWS Software working as a service over the internet encouraging effortless cooperation and ease of access.
- **Automation of Updates:** SaaS providers manage the handling of software maintenance with automatic latest updates ensuring users gain experience with the latest features and security patches.
- **Cost Efficiency:** SaaS acts as a cost-effective solution by reducing the overhead of IT support by eliminating the need for individual software licenses.

Top Leading Cloud Computing Companies

The following tables show the top leading cloud computing companies along with key details about their cloud services:

Company	Cloud Service Name	Key Offerings
1. Amazon	AWS (Amazon Web Services)	Compute, Storage, AI/ML, Databases, Networking
2. Microsoft	Azure	Cloud computing, AI, Analytics, Hybrid Cloud
3. Google	Google Cloud Platform (GCP)	AI/ML, Big Data, Kubernetes, Cloud Storage
4. Alibaba	Alibaba Cloud	IaaS, AI, Big Data, Cloud Security, CDN
5. Oracle	Oracle Cloud	Enterprise Cloud, Databases, SaaS, PaaS
6. IBM	IBM Cloud	AI, Quantum Computing, Hybrid Cloud, Security
7. Salesforce	Salesforce Cloud	CRM, SaaS, AI, Analytics
8. Tencent	Tencent Cloud	AI, Gaming Cloud, IoT, Big Data

Cloud Security

[Cloud security](#) recommended to measures and practices designed to protect data, applications, and infrastructure in cloud computing environments. The following are some of the best practices of cloud security:

1. Data Encryption

Encryption is essential for securing data stored in the cloud. It ensures that data remains unreadable to unauthorized users even if it is intercepted.

2. Access Control

Implementing strict access controls and authentication mechanisms helps ensure that only authorized users can access sensitive data and resources in the cloud.

3. Multi-Factor Authentication (MFA)

[MFA](#) adds an extra layer of security by requiring users to provide multiple forms of verification, such as passwords, biometrics, or security tokens, before gaining access to cloud services.

Use Cases Of Cloud Computing

Cloud computing is used across all industries for various applications:

- **Scalable Websites and Applications:** Host applications that can handle millions of users.

- **Data Storage, Backup, and Recovery:** Cost-effectively store vast amounts of data and ensure it can be recovered in case of a disaster.
- **Big Data Analytics:** Process and derive insights from enormous datasets efficiently.
- **Software Development and Testing:** Create and dismantle development environments quickly.
- **Artificial Intelligence and Machine Learning:** Access powerful GPU resources to train complex AI models.

NIST CLOUD COMPUTING

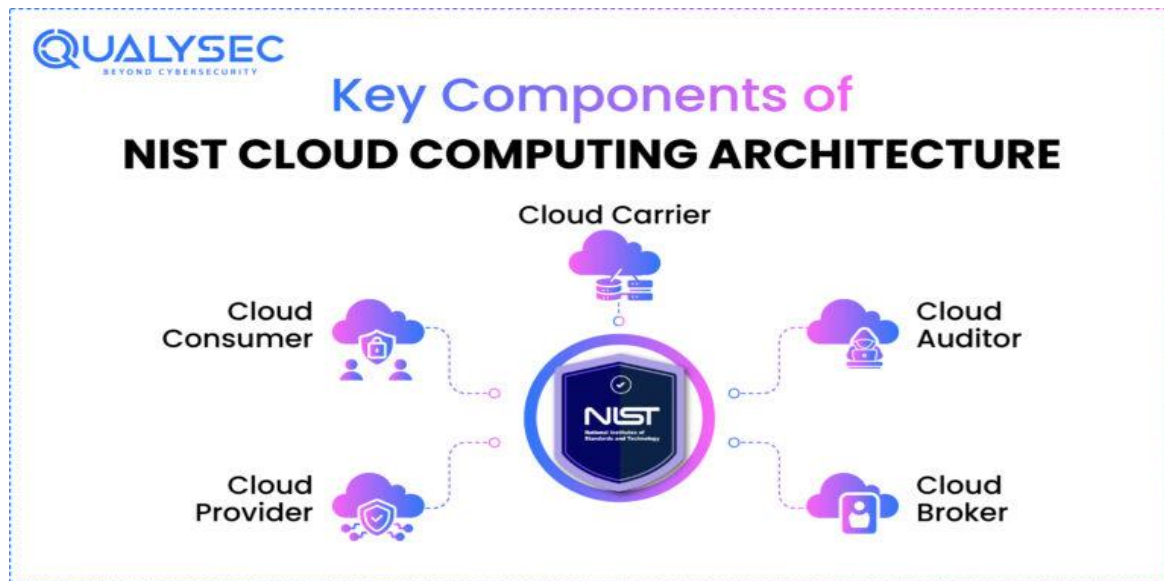
The National Institute of Standards and Technology (NIST) provides foundational definitions, characteristics, and reference architectures for cloud computing, defining it as a model for on-demand network access to shared, configurable resources, with key components like [IaaS](#), [PaaS](#), [SaaS](#), and models like public, private, hybrid. NIST's framework clarifies roles (consumer, provider, etc.) and essential traits (self-service, broad access, elasticity) to promote secure, standardized, and interoperable cloud environments, helping organizations understand cloud risks and governance.

Key NIST Cloud Concepts:

- **Definition:** A model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (networks, servers, storage, applications, services) that can be rapidly provisioned and released with minimal management effort.
- **Essential Characteristics:**
 - **On-Demand Self-Service:** Users can provision resources automatically.
 - **Broad Network Access:** Available over the network via standard mechanisms.
 - **Resource Pooling:** Provider resources serve multiple consumers.
 - **Rapid Elasticity:** Scalable up or down quickly.
 - **Measured Service:** Usage is monitored and billed (pay-as-you-go).
- **Service Models:**
 - **Infrastructure as a Service (IaaS):** Access to fundamental computing resources (VMs, storage).
 - **Platform as a Service (PaaS):** Platform for developing, running applications (e.g., operating systems, databases).
 - **Software as a Service (SaaS):** Ready-to-use applications over the internet (e.g., email, CRM).
- **Deployment Models:**
 - **Public Cloud:** Available to the general public.
 - **Private Cloud:** Exclusive use by a single organization.
 - **Community Cloud:** Shared by several organizations with common concerns.
 - **Hybrid Cloud:** Combination of two or more distinct cloud infrastructures.
- **Reference Architecture (Actors):**
 - **Cloud Consumer:** Uses cloud services.
 - **Cloud Provider:** Offers services.
 - **Cloud Carrier:** Provides network connectivity.

- **Cloud Broker:** Manages service usage, performance.
- **Cloud Auditor:** Assesses controls and compliance.

NIST's work, provides a common language and framework for understanding, securing, and managing cloud computing across federal agencies and industry.



Cloud Deployment Models

- Cloud Computing has now become an essential part of modern businesses, offering flexibility, scalability, and cost-effective solutions.
- But Selecting the most appropriate cloud deployment model is essential to utilize the complete potential of cloud services.
- Whether you're a small business or a large enterprise, choosing the right cloud model can affect your security, scalability, and operational efficiency.

What is a Cloud Deployment Model?

- A cloud deployment model fundamentally determines where the infrastructure for your deployment exists and who owns and controls that infrastructure. It also figures out the nature and intention of the cloud.
- The initial step for any business seeking to embrace cloud services is to be familiar with the deployment models that are available.
- When these are familiar, an improved choice can be made as to which avenues the company should pursue.
- Every model will provide its strengths and weaknesses in governance, scalability, security, flexibility, cost, and management
- The cloud deployment model identifies the specific type of cloud environment based on ownership, scale, and access, as well as the cloud's nature and purpose.

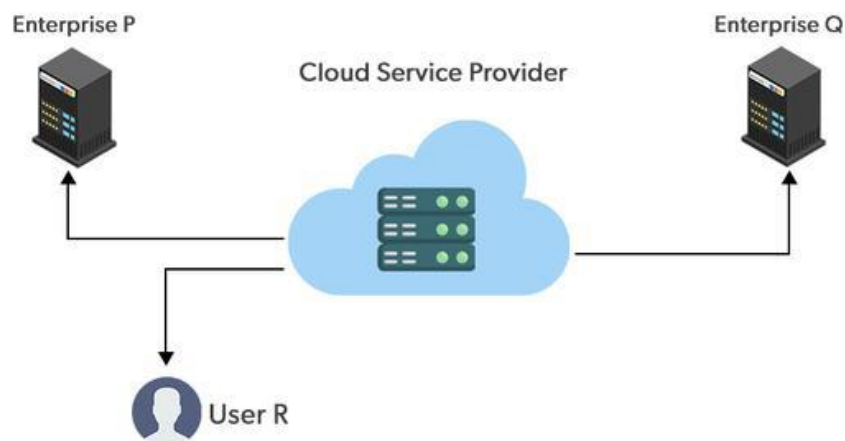
- The location of the servers you're utilizing and who controls them are defined by a cloud deployment model.
- It specifies how your cloud infrastructure will look, what you can change, and whether you will be given services or will have to create everything yourself.
- Relationships between the infrastructure and your users are also defined by cloud deployment types.

Different types of cloud computing deployment models are described below.

- Public Cloud
- Private Cloud
- Hybrid Cloud
- Community Cloud
- Multi-Cloud

Public Cloud

The public cloud makes it possible for anybody to access systems and services. The public cloud may be less secure as it is open to everyone. The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups. The infrastructure in this cloud model is owned by the entity that delivers the cloud services, not by the consumer. It is a type of cloud hosting that allows customers and users to easily access systems and services. This form of cloud computing is an excellent example of cloud hosting, in which service providers supply services to a variety of customers. In this arrangement, storage backup and retrieval services are given for free, as a subscription, or on a per-user basis. For example, GoogleAppEngine



Advantages of the Public Cloud Model

- **Minimal Investment:** Because it is a pay-per-use service, there is no substantial upfront fee, making it excellent for enterprises that require immediate access to resources.
- **No setup cost:** The entire infrastructure is fully subsidized by the cloud service providers, thus there is no need to set up any hardware.
- **Infrastructure Management is not required:** Using the public cloud does not necessitate infrastructure management.
- **No maintenance:** The maintenance work is done by the service provider (not users).
- **Dynamic Scalability:** To fulfill your company's needs, on-demand resources are accessible.

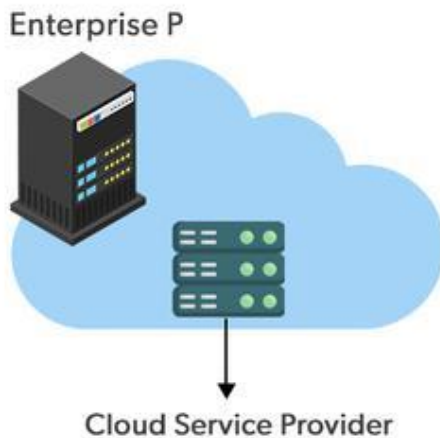
Disadvantages of the Public Cloud Model

- **Less secure:** Public cloud is less secure as resources are public so there is no guarantee of high-level security.
- **Low customization:** It is accessed by many public so it can't be customized according to personal requirements.

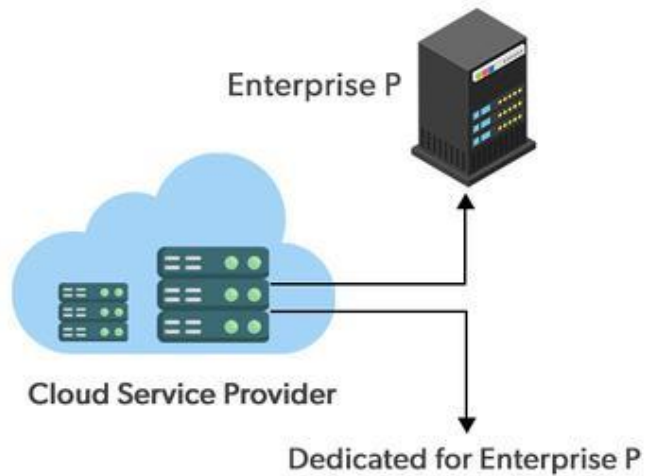
Private Cloud

The private cloud deployment model is the exact opposite of the public cloud deployment model. It's a one-on-one environment for a single user (customer). There is no need to share your hardware with anyone else. The distinction between [private and public clouds](#) is in how you handle all of the hardware. It is also called the "internal cloud" & it refers to the ability to access systems and services within a given border or organization. The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization's IT department. The private cloud gives greater flexibility of control over cloud resources.

On premise Private cloud



Externally hosted Private cloud



Advantages of the Private Cloud Model

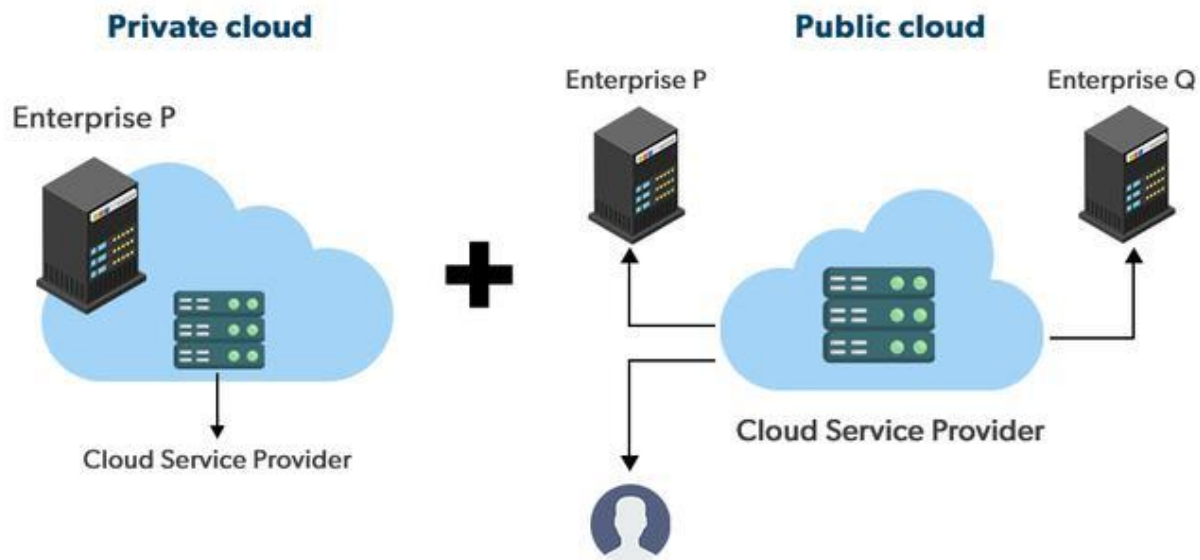
- **Better Control:** You are the sole owner of the property. You gain complete command over service integration, IT operations, policies, and user behavior.
- **Data Security and Privacy:** It's suitable for storing corporate information to which only authorized staff have access. By segmenting resources within the same infrastructure, improved access and security can be achieved.
- **Supports Legacy Systems:** This approach is designed to work with legacy systems that are unable to access the public cloud.
- **Customization:** Unlike a public cloud deployment, a private cloud allows a company to tailor its solution to meet its specific needs.

Disadvantages of the Private Cloud Model

- **Less scalable:** Private clouds are scaled within a certain range as there is less number of clients.
- **Costly:** Private clouds are more costly as they provide personalized facilities.

Hybrid Cloud

By bridging the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds. With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud's cost savings. Organizations can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs.



Advantages of the Hybrid Cloud Model

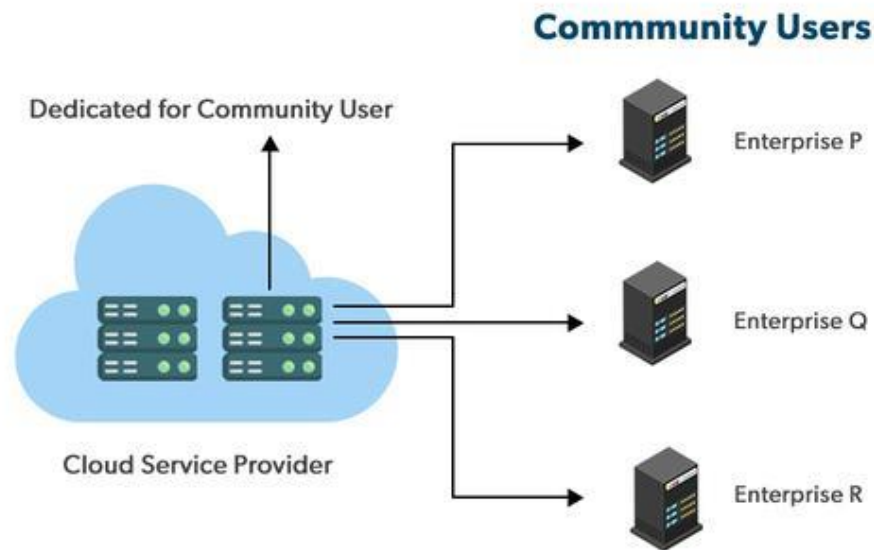
- **Flexibility and control:** Businesses with more flexibility can design personalized solutions that meet their particular needs.
- **Cost:** Because public clouds provide scalability, you'll only be responsible for paying for the extra capacity if you require it.
- **Security:** Because data is properly separated, the chances of data theft by attackers are considerably reduced.

Disadvantages of the Hybrid Cloud Model

- **Difficult to manage:** Hybrid clouds are difficult to manage as it is a combination of both public and private cloud. So, it is complex.
- **Slow data transmission:** Data transmission in the hybrid cloud takes place through the public cloud so latency occurs.

Community Cloud

It allows systems and services to be accessible by a group of organizations. It is a distributed system that is created by integrating the services of different clouds to address the specific needs of a community, industry, or business. The infrastructure of the community could be shared between the organization which has shared concerns or tasks. It is generally managed by a third party or by the combination of one or more organizations in the community.



Advantages of the Community Cloud Model

- **Cost Effective:** It is cost-effective because the cloud is shared by multiple organizations or communities.
- **Security:** Community cloud provides better security.
- **Shared resources:** It allows you to share resources, infrastructure, etc. with multiple organizations.
- **Collaboration and data sharing:** It is suitable for both collaboration and data sharing.

Disadvantages of the Community Cloud Model

- **Limited Scalability:** Community cloud is relatively less scalable as many organizations share the same resources according to their collaborative interests.
- **Rigid in customization:** As the data and resources are shared among different organizations according to their mutual interests if an organization wants some changes according to their needs they cannot do so because it will have an impact on other organizations.

Cloud Deployment Models Comparison

The overall Analysis of these models with respect to different factors is described below.

Factors	Public Cloud	Private Cloud	Community Cloud	Hybrid Cloud
Initial Setup	Easy	Complex, requires a professional team to setup	Complex, requires a professional team to setup	Complex, requires a professional team to setup
Scalability and Flexibility	High	High	Fixed	High
Cost-Comparison	Cost-Effective	Costly	Distributed cost among members	Between public and private cloud
Reliability	Low	Low	High	High
Data Security	Low	High	High	High
Data Privacy	Low	High	High	High

Models of Cloud Computing

Cloud Computing helps in rendering several services according to roles, companies, etc. Cloud computing models are explained below.

- Infrastructure as a service (IaaS)
- Platform as a service (PaaS)
- Software as a service (SaaS)

1. Infrastructure as a service (IaaS)

[Infrastructure as a Service \(IaaS\)](#) helps in delivering computer infrastructure on an external basis for supporting operations. Generally, IaaS provides services to networking equipment, devices, databases, and web servers.

Infrastructure as a Service (IaaS) helps large organizations, and large enterprises in managing and building their IT platforms. This infrastructure is flexible according to the needs of the client.

Advantages of IaaS

- IaaS is cost-effective as it eliminates capital expenses.
- IaaS cloud provider provides better security than any other software.
- IaaS provides remote access.

Disadvantages of IaaS

- In IaaS, users have to secure their own data and applications.
- Cloud computing is not accessible in some regions of the World.

2. Platform as a service (PaaS)

Platform as a Service (PaaS) is a type of cloud computing that helps developers to build applications and services over the Internet by providing them with a platform.

PaaS helps in maintaining control over their business applications.

Advantages of PaaS

- PaaS is simple and very much convenient for the user as it can be accessed via a web browser.
- PaaS has the capabilities to efficiently manage the lifecycle.

Disadvantages of PaaS

- PaaS has limited control over infrastructure as they have less control over the environment and are not able to make some customizations.
- PaaS has a high dependence on the provider.

3. Software as a service (SaaS)

Software as a Service (SaaS) is a type of cloud computing model that is the work of delivering services and applications over the Internet. The SaaS applications are called Web-Based Software or Hosted Software.

SaaS has around 60 percent of cloud solutions and due to this, it is mostly preferred by companies.

Advantages of SaaS

- SaaS can access app data from anywhere on the Internet.
- SaaS provides easy access to features and services.

Disadvantages of SaaS

- SaaS solutions have limited customization, which means they have some restrictions within the platform.
- SaaS has little control over the data of the user.

- SaaS are generally cloud-based, they require a stable internet connection for proper working.

Characteristics of Cloud Computing

There are many characteristics of [Cloud Computing](#) here are few of them :

1. **On-demand self-services:** The Cloud computing services does not require any human administrators, user themselves are able to provision, monitor and manage computing resources as needed.
2. **Broad network access:** The Computing services are generally provided over standard networks and heterogeneous devices.
3. **Rapid elasticity:** The Computing services should have IT resources that are able to scale out and in quickly and on a need basis. Whenever the user require services it is provided to him and it is scale out as soon as its requirement gets over.
4. **Resource pooling:** The IT resource (e.g., networks, servers, storage, applications, and services) present are shared across multiple applications and occupant in an uncommitted manner. Multiple clients are provided service from a same physical resource.
5. **Measured service:** The resource utilization is tracked for each application and occupant, it will provide both the user and the resource provider with an account of what has been used. This is done for various reasons like monitoring billing and effective use of resource.
6. **Multi-tenancy:** Cloud computing providers can support multiple tenants (users or organizations) on a single set of shared resources.
7. **Virtualization:** Cloud computing providers use virtualization technology to abstract underlying hardware resources and present them as logical resources to users.
8. **Resilient computing:** Cloud computing services are typically designed with redundancy and fault tolerance in mind, which ensures high availability and reliability.
9. **Flexible pricing models:** Cloud providers offer a variety of pricing models, including pay-per-use, subscription-based, and spot pricing, allowing users to choose the option that best suits their needs.
10. **Security:** Cloud providers invest heavily in security measures to protect their users' data and ensure the privacy of sensitive information.
11. **Automation:** Cloud computing services are often highly automated, allowing users to deploy and manage resources with minimal manual intervention.

12. **Sustainability:** Cloud providers are increasingly focused on sustainable practices, such as energy-efficient data centers and the use of renewable energy sources, to reduce their environmental impact.

Cloud Computing Stack

The cloud computing stack is a layered architecture comprising SaaS, PaaS, and IaaS, designed to deliver scalable, virtualized IT resources over the internet. It operates as a service-oriented model, enabling on-demand, metered access to infrastructure, development tools, and ready-to-use software applications.

Core Layers of the Cloud Computing Stack

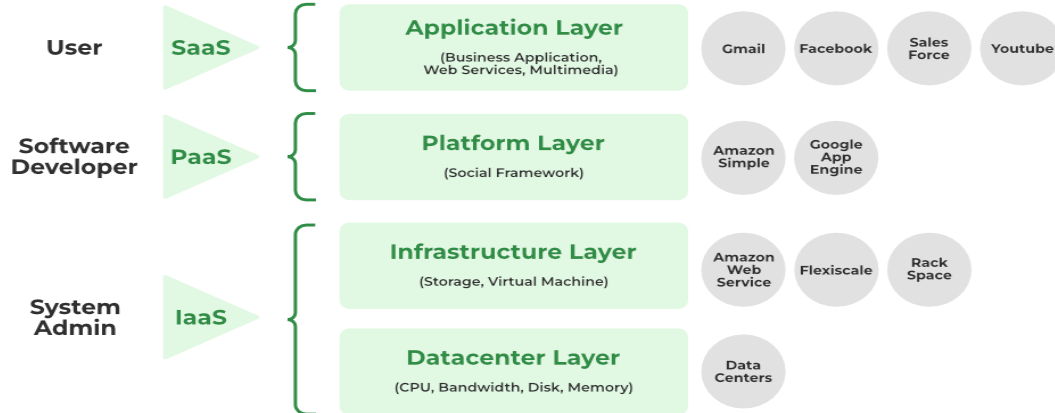
- **Software as a Service (SaaS):** The top layer, delivering fully managed applications to end-users via web browsers or apps (e.g., Salesforce, Microsoft 365).
- **Platform as a Service (PaaS):** The middle layer, providing developers with tools, databases, and operating systems to build and deploy applications without managing underlying infrastructure.
- **Infrastructure as a Service (IaaS):** The foundational layer, offering virtualized computing resources like servers, storage, and networking over the internet, allowing full control over OS and applications.

Key Characteristics and Components

- **Virtualization:** The core enabling technology that abstracts physical hardware into manageable, flexible, and scalable virtual machines.
- **Deployment Models:** The stack can be implemented in Public, Private, Hybrid, or Community clouds.
- **Resource Management:** Features rapid elasticity for scaling and metered (pay-per-use) billing.
- **Components:** A typical stack includes load balancers, firewalls, and security certificates.

The stack is often compared to a cake, where each layer represents a different level of control and functionality for the user.

Cloud Computing Layers



OpenStack

- It is a free open standard cloud computing platform that first came into existence on July 21' 2010.
- It was a joint project of Rackspace Hosting and NASA to make cloud computing more ubiquitous in nature.
- It is deployed as Infrastructure-as-a-service(IaaS) in both public and private clouds where virtual resources are made available to the users.
- The software platform contains interrelated components that control multi-vendor hardware pools of processing, storage, networking resources through a data center.
- In OpenStack, the tools which are used to build this platform are referred to as "projects". These projects handle a large number of services including computing, networking, and storage services.
- Unlike virtualization, in which resources such as RAM, CPU, etc are abstracted from the hardware using hypervisors,
- OpenStack uses a number of APIs to abstract those resources so that users and the administrators are able to directly interact with the cloud services.

OpenStack components

Apart from various projects which constitute the OpenStack platform, there are **nine major** services namely Nova, Neutron, Swift, Cinder, Keystone, Horizon, Ceilometer, and Heat. Here is the basic definition of all the components which will give us a basic idea about these components.

1. **Nova (compute service):** It manages the compute resources like creating, deleting, and handling the scheduling. It can be seen as a program dedicated to the automation of resources that are responsible for the virtualization of services and high-performance computing.
2. **Neutron (networking service):** It is responsible for connecting all the networks across OpenStack. It is an API driven service that manages all networks and IP addresses.
3. **Swift (object storage):** It is an object storage service with high fault tolerance capabilities and it used to retrieve unstructured data objects with the help of Restful API. Being a distributed platform, it is also used to provide redundant storage within servers that are clustered together. It is able to successfully manage petabytes of data.
4. **Cinder (block storage):** It is responsible for providing persistent block storage that is made accessible using an API (self- service). Consequently, it allows users to define and manage the amount of cloud storage required.
5. **Keystone (identity service provider):** It is responsible for all types of authentications and authorizations in the OpenStack services. It is a directory-based service that uses a central repository to map the correct services with the correct user.
6. **Glance (image service provider):** It is responsible for registering, storing, and retrieving virtual disk images from the complete network. These images are stored in a wide range of back-end systems.
7. **Horizon (dashboard):** It is responsible for providing a web-based interface for OpenStack services. It is used to manage, provision, and monitor cloud resources.
8. **Ceilometer (telemetry):** It is responsible for metering and billing of services used. Also, it is used to generate alarms when a certain threshold is exceeded.
9. **Heat (orchestration):** It is used for on-demand service provisioning with auto-scaling of cloud resources. It works in coordination with the ceilometer.

These are the services around which this platform revolves around. These services individually handle storage, compute, networking, identity, etc. These services are the base on which the rest of the projects rely on and are able to orchestrate services, allow bare-metal provisioning, handle dashboards, etc.

Features of OpenStack

- **Modular architecture:** OpenStack is designed with a modular architecture that enables users to deploy only the components they need. This makes it easier to customize and scale the platform to meet specific business requirements.
- **Multi-tenancy support:** OpenStack provides multi-tenancy support, which enables multiple users to access the same cloud infrastructure while maintaining security and isolation between them. This is particularly important for cloud service providers who need to offer services to multiple customers.
- **Open-source software:** OpenStack is an open-source software platform that is free to use and modify. This enables users to customize the platform to meet their specific requirements, without the need for expensive proprietary software licenses.
- **Distributed architecture:** OpenStack is designed with a distributed architecture that enables users to scale their cloud infrastructure horizontally across multiple physical servers. This makes it easier to handle large workloads and improve system performance.

- **API-driven:** OpenStack is API-driven, which means that all components can be accessed and controlled through a set of APIs. This makes it easier to automate and integrate with other tools and services.
- **Comprehensive dashboard:** OpenStack provides a comprehensive dashboard that enables users to manage their cloud infrastructure and resources through a user-friendly web interface. This makes it easier to monitor and manage cloud resources without the need for specialized technical skills.
- **Resource pooling:** OpenStack enables users to pool computing, storage, and networking resources, which can be dynamically allocated and de-allocated based on demand. This enables users to optimize resource utilization and reduce waste.

Advantages of using OpenStack

- It boosts rapid provisioning of resources due to which orchestration and scaling up and down of resources becomes easy.
- Deployment of applications using OpenStack does not consume a large amount of time.
- Since resources are scalable therefore they are used more wisely and efficiently.
- The regulatory compliances associated with its usage are manageable.

Disadvantages of using OpenStack

- OpenStack is not very robust when orchestration is considered.
- Even today, the APIs provided and supported by OpenStack are not compatible with many of the hybrid cloud providers, thus integrating solutions becomes difficult.
- Like all cloud service providers OpenStack services also come with the risk of security breaches.